

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV Examination, 2022-23

Paper: BCH-226: Business Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Attempt all questions. All questions carry equal marks (14 marks each).***Question 1.**

“Index numbers are a specialised type of averages.” Comment on this statement. Also state briefly the various points to be considered in the construction of index numbers. [14]

OR

The following are the average daily wages and price indices of a group of industrial workers:

Year	Average Daily Wages (Rs.)	Price Index
1999	80	100
2000	108	120
2001	125	125
2002	147	140
2003	216	180
2004	230	200

(a) Compute the index number of real wages. [8]

(b) In which year do the workers have maximum buying power? [6]

Question 2.

What do you mean by “Cost of Living Index”? State its kinds. Explain any one of its kinds with the help of imaginary figures. [14]

OR

Compute Fisher’s Ideal Index Number from the following data:

Items	Base Year		Current Year	
	Price	Quantity	Price	Quantity
A	10	40	12	50
B	12	25	15	20
C	15	10	20	12
D	20	5	30	2

Also examine if it satisfies the Time Reversal Test and Factor Reversal Test. [14]

Question 3.

What are the components of a Time Series? Explain the significance of moving averages in analyzing time series and point out its limitations. [14]

OR

The following data relates to the sales of Dinkar Limited:

Year	2018	2019	2020	2021	2022
Sales (Rs. in lakhs)	80	120	160	360	280

- (a) Fit a straight line trend by the method of least squares and tabulate the trend values. [6]
- (b) Estimate the sales for the year 2023. [4]
- (c) Measure the short-term fluctuations. [4]

Question 4.

What are the various methods of measuring seasonal variation in a time series? Discuss any one method in detail. [14]

OR

Calculate the seasonal index by the link relative method from the following link relatives:

Quarter	2018	2019	2020	2021	2022
I	—	80	88	80	83
II	120	117	129	125	117
III	133	113	111	115	120
IV	83	89	93	96	79

[14]

Question 5.

What do you understand by “Statistical Quality Control”? Discuss its techniques and advantages. [14]

OR

The mean and range of 10 random samples of size 4 are as follows:

Sample No.	1	2	3	4	5	6	7	8	9	10
Sample Mean ($\bar{x}$)	494	508	500	582	552	538	514	614	600	558
Sample Range (R)	95	128	100	91	68	65	148	28	37	80

Determine the control limits for the $\bar{x}$ chart. Draw the three lines of the mean chart on graph paper. Also determine whether the process is within control or not? You are given that $d_2 = 2.059$ for a sample size of 4. [14]

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